

Video 4.1.1

Homogeneous systems of linear equation.

These are easy to define. A system of linear equations is said to be **homogeneous** if the constant term in each equation is zero.

For example:

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ -x_1 + 2x_2 - 3x_3 = 0 \\ 2x_1 + 4x_2 + 6x_3 = 0 \end{cases}$$

In other words, a system is homogeneous if the corresponding augmented matrix is all zero to the right of the partition:

$$\left[\begin{array}{ccc|c} * & \cdots & * & 0 \\ \vdots & \vdots & \vdots & \vdots \\ * & \cdots & * & 0 \end{array} \right]$$

All zeros.

Here is the important special fact about homogeneous linear systems.

Fact (Existence of a trivial solution to a homogeneous system.)

A homogeneous system of linear equations is **always** solved by setting every variable to zero. ** guaranteed solution -> the origin.*

In other words: the point $(0, \dots, 0) \in \mathbb{R}^n$ is **always** a solution to the system (if the system has n variables).

This solution is called the **trivial solution**. *-> the origin.*

Any other solution is called a **non-trivial solution**.

Geometric interpretation.

Question: What is a geometric interpretation of a homogeneous system?

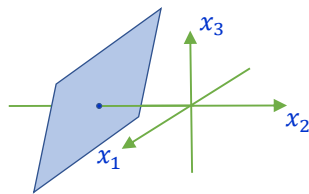
Answer:

If we think of the set of solutions as some subset of $\mathbb{R}^n$ then the set of solutions **must include the origin**.

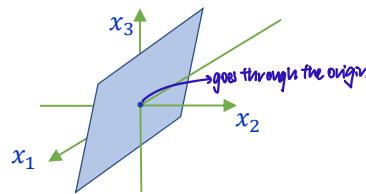
For example: The set of solutions to an equation

$$ax_1 + bx_2 + cx_3 = d$$

is a general plane in $\mathbb{R}^3$. In the case that the equation is homogeneous $d = 0$ then the plane **must go through the origin**.



The inhomogeneous case: $d \neq 0$.



The homogeneous case: $d = 0$.

Counting solutions to homogeneous systems

Because a homogeneous system always contains the trivial solution, it is **always consistent**.

So the possibilities for the set of solutions are simpler than in general.

If we restrict to homogeneous systems, the usual three possibilities becomes **only two possibilities**:

- **Possibility #1.** The system only has the trivial solution. ↗ solution
- **Possibility #2.** The system has infinitely many solutions, including the trivial solution.

Example:

Video 2.1.2

Before analyzing things generally, let's try and understand geometrically the possibilities for a homogeneous system of 3 linear equations in 3 variables.

$$\begin{aligned}a_{11}x + a_{12}y + a_{13}z &= 0 \\a_{21}x + a_{22}y + a_{23}z &= 0 \\a_{31}x + a_{32}y + a_{33}z &= 0\end{aligned}$$

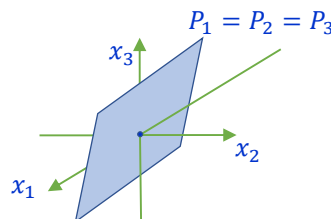
Discussion:

- Assuming each equation is not zero then the solutions to each equation separately form a plane through the origin. Denote the planes P_1 , P_2 , and P_3 .
- The set of solutions to the system are equivalent to the points which lie in all three of the three planes: $P_1 \cap P_2 \cap P_3$.
- Obviously each plane contains the origin, so the origin will always be a point lying in the intersection: $(0,0,0) \in P_1 \cap P_2 \cap P_3$.
- We need to understand using geometric intuition: What are the geometric possibilities for the other points in $P_1 \cap P_2 \cap P_3$?

Discussion (continued):

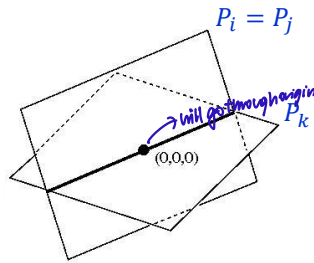
- The first possibility is that all three planes are exactly the same plane through the origin. In that case their intersection is the same plane. This case gives an infinite number of solutions. We expect two free parameters in the general solution.

↳ because the plane is 2-dimensional

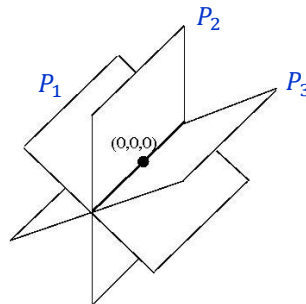


Discussion (continued):

- The second possibility is that two of the planes, say P_i and P_j , are exactly the same and the third P_k is different. In this case the three planes will definitely intersect in a line. So in this case we expect infinitely many solutions and one free parameter.

Discussion (continued):

- The third possibility is that all three planes are different, but that all three planes meet along the same line. In this case we expect infinitely many solutions and one free parameter.



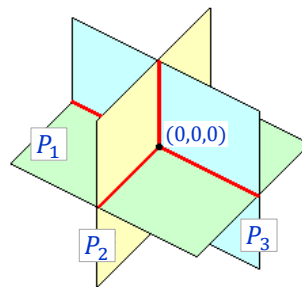
Discussion (continued):

→ most common scenario

- The final possibility is that all three planes are different, and their only point of intersection is the origin. An obvious example of this is the three coordinate planes, which corresponds to the

system $\begin{cases} x = 0 \\ y = 0 \\ z = 0 \end{cases}$. In this case the system **only has the trivial**

solution.



Video 4.1.3

Counting solutions to homogeneous systems (cont.)

A general analysis:

Let's quickly analyse the possible number of solutions for a homogeneous system according to the number of equations versus the number of variables:

1. The case $(\# \text{ of equations}) < (\# \text{ variables})$.
2. The case $(\# \text{ of equations}) = (\# \text{ variables})$.
3. The case $(\# \text{ of equations}) > (\# \text{ variables})$.

*will never have an inconsistent system.

Case 1: A homogeneous system with (# of equations) < (# variables)

This case will always have an *infinite number of solutions, including the trivial solution.*

Why?

The result of Gaussian elimination will produce an augmented matrix which is consistent (because the system is homogeneous) and has more variables than rows. Thus there will be some column without a leading entry, which *implies a free parameter.*

A simple example is the plane through the origin:

$$x + y + z = 0.$$

Case 2: A homogeneous system with (# of equations) = (# variables)

This case can either

1. have an *infinite number of solutions including the trivial solution.*
2. have *only the trivial solution.*

Why?

Gaussian elimination can reduce the number of rows, so the result of Gaussian elimination in this case will produce an augmented matrix *which is consistent* and either (i) has more variables than rows or (ii) has the same number of variables as rows. In the first case there will be infinitely many solutions, and in the second case there will be exactly one solution.

For example:

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ x_2 + x_3 &= 0 \\ x_1 + 2x_2 + 2x_3 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ x_2 + x_3 &= 0 \\ x_3 &= 0 \end{aligned}$$

Infinitely many solutions. (Check).

Only the trivial solution.

→ In row echelon form there will be leading entries in every column, so no free parameters.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \text{infinitely many solutions.}$$

Case 3: A homogeneous system with (# of equations) > (# variables)

This case can either

1. have an *infinite number of solutions including the trivial solution.*
2. have *only the trivial solution.*

Why?

Just like in case 2, Gaussian elimination will produce an augmented matrix which is **consistent** and either (i) has more variables than rows or (ii) has the same number of variables as rows. (Check you understand why there cannot be more rows than variables.) In the first case there will be infinitely many solutions, and in the second case there will be exactly one solution.

→ in row echelon form, leading entries move strictly to the right, so it won't ever be the case where there are more rows than variables.

For example:

$$\begin{aligned}x_1 + x_2 + x_3 &= 0 \\x_2 + x_3 &= 0 \\x_1 + 2x_2 + 2x_3 &= 0 \\2x_1 + 4x_2 + 4x_3 &= 0\end{aligned}$$

*Infinitely many solutions. Check.
row echelon form has 2 eqn and 3 variables.*

$$\begin{aligned}x_1 + x_2 + x_3 &= 0 \\x_2 + x_3 &= 0 \\x_3 &= 0 \\any\ linear\ expression &= 0\end{aligned}$$

Only the trivial solution. Why?

first 3 eqn already show that the only possible soln is the origin.

Summary:

